

Differentiable functions – 6

Differential inequalities and comparison

Homework

1 сентября 2026 года

Theoretical minimum. Gronwall's lemma in the integral and the differential form; the integrating factor $e^{-\int b}$; comparison theorems for first and second order equations; the Wronskian and the interlacing of zeros; Sturm's comparison and oscillation theorems; the Riccati substitution $u = y'/y$; a priori bounds and blow-up.

1. Let f be differentiable with $f'(x) \leq \alpha f(x) + \beta$ and $f(0) = f_0$. Find the best possible upper bound for $f(x)$, $x \geq 0$, and show that it is attained.
2. Prove that the Wronskian of two solutions of $y'' + py' + qy = 0$ satisfies $W' = -pW$ (Abel's identity), and deduce that two solutions are independent at one point if and only if they are independent everywhere.
3. Prove that between two consecutive zeros of a nontrivial solution of $y'' + qy = 0$ there is exactly one zero of every independent solution.
4. Let $y'' + q(x)y = 0$ with $0 < m^2 \leq q(x) \leq M^2$. Prove that the distance between consecutive zeros of a nontrivial solution lies between $\frac{\pi}{M}$ and $\frac{\pi}{m}$.
5. Let u be non-negative with $u' \leq u^2$ and $u(0) = u_0 > 0$. Prove that u blows up no later than at $t = \frac{1}{u_0}$, and give the sharp example.
6. Let f be differentiable on $\mathbb{R}$ with $|f'(x)| \leq |f(x)|$ and $f(x_0) = 0$ for some x_0 . Prove that $f \equiv 0$.
7. Prove the Bihari–LaSalle nonlinear version of Gronwall's lemma: $u(t) \leq a + \int_0^t b(s)\omega(u(s)) ds$ with ω increasing implies an explicit bound through Ω^{-1} , where $\Omega(x) = \int_1^x \frac{ds}{\omega(s)}$.

8. Let f satisfy $f'' + f \geq 0$ on $[0, \pi]$ with $f(0) = f(\pi) = 0$. Prove that $f \leq 0$ on $[0, \pi]$, and explain how this is the second-order maximum principle.
9. Prove that a solution of $y' = y^2 - x$ defined on all of $\mathbb{R}$ is unique if it exists, and describe the qualitative behaviour of solutions using the comparison theorem.
10. Prove that if f is twice differentiable with $f + f'' \geq 0$ and $f(0) = f(\pi) = 0$, then $\int_0^\pi f(x) \sin x \, dx \geq 0$, and deduce a positivity statement for the Green's function of the operator $-\frac{d^2}{dx^2} - 1$.